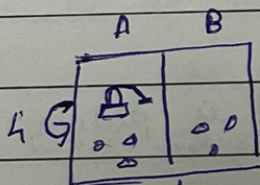


2. Implement vacuum-cleaning Agent (Simple reflex-agent) function

1. Initialize two rooms A and B with clean or dirty
2. Position vacuum cleaner at room A
3. Check if A is dirty
if dirty perform 'clean'
4. Check if A is clean
move to room B
5. Check if B is dirty
if dirty, perform 'clean' and move to A
6. Continue the same process
7. If both A & B are clean stop

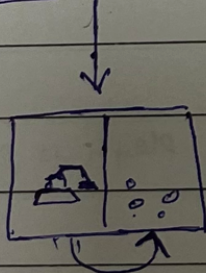
It will have $n \times 2^n$ states

Diagram

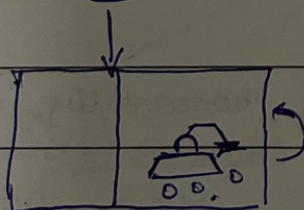


$\{ Loc(A), Dirt(A), Dirt(B) \}$

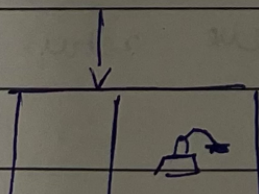
~~$\{ Loc(A) \}$~~



$\{ Loc(A), clean(A), Dirt(B) \}$



$\{ Loc(B), clean(A), Dirt(B) \}$



$\{ Loc(B), clean(A), clean(B) \}$

Pseudocode:-

```
function simple_reflex_agent returns an action
  if status == 'Dirty' then return pick the dirt
  else if location = A then return right
  { else if location = B then return left
```

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